

Variational inference

Choose a simple family of distributions and tune its parameters until one member sits as close to the posterior as the ELBO can tell.

THE QUESTION

The posterior $p(\theta | D) = p(D | \theta)p(\theta)/p(D)$ needs the evidence $p(D)$, an integral we usually cannot do. Can optimisation find the closest simple distribution instead?

THE INTUITION

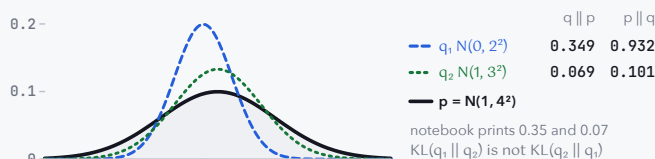
Pick a family $q_\psi(\theta)$, say Gaussians, and minimise $\text{KL}(q_\psi \| p(\theta | D))$. The unknown $\log p(D)$ is a constant, so this equals maximising the ELBO, which needs only the joint $p(D, \theta)$ and draws from q .

KEY TAKEAWAY

$\log p(D) = \text{ELBO}(q) + \text{KL}(q \| p(\theta | D))$. Raising the ELBO pushes q toward the posterior, and the gap left is the KL. This KL seeks one mode and underestimates spread.

$p(D, \theta) \rightarrow$ family $q_\psi(\theta) \rightarrow$ draw $\theta = \mu + \sigma \epsilon \rightarrow$ estimate the ELBO $\rightarrow$ gradient step on $\psi \rightarrow q \approx p(\theta | D)$

1 KL measures how far q is from p



Adam on $\text{KL}(q \| p)$ from $N(4, 0.5^2)$: (1.06, 3.15) after 100 steps, (1.00, 4.00) after 300

$$\text{KL}(q \| p) = \int q(\theta) \log \frac{q(\theta)}{p(\theta)} d\theta = \mathbb{E}_q[\log q(\theta) - \log p(\theta)]$$

It is ≥ 0 , zero only when $q = p$, and not symmetric.

For two Gaussians (the slides' exercise, using $\mathbb{E}_q[\theta^2] = \sigma_q^2 + \mu_q^2$)

$$\text{KL}(q \| p) = \log \frac{\sigma_p}{\sigma_q} + \frac{\sigma_q^2 + (\mu_q - \mu_p)^2}{2\sigma_p^2} - \frac{1}{2}$$

Minimising it over (μ_q, σ_q) recovers p . With no formula, average $\log q - \log p$ over draws from q ($q = \mathcal{N}(3, 1)$ against the notebook's mixture: 9.46 quoted, 9.489 on a grid). More on KL: the information theory sheet.

2 $\log p(D) = \text{ELBO} + \text{KL}(q \| \text{posterior})$



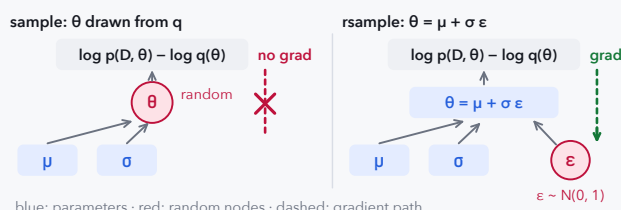
Put Bayes' rule $p(\theta | D) = p(D, \theta)/p(D)$ inside the KL: $\text{KL}(q \| p(\theta | D)) = \mathbb{E}_q[\log q(\theta) - \log p(D, \theta)] + \log p(D)$

$\log p(D)$ does not depend on θ , so it leaves the expectation. Rearranged:

$$\log p(D) = \underbrace{\mathbb{E}_q[\log p(D, \theta) - \log q(\theta)]}_{\text{ELBO}(q)} + \underbrace{\text{KL}(q \| p(\theta | D))}_{\geq 0}$$

The ELBO needs only the joint and bounds the log evidence from below; with $\log p(D)$ fixed, each nat it gains is a nat of KL lost. Coin: Beta(5, 3) prior, 17 heads in 30, $\log p(D) = -21.397$ (the notebook's loss at the true posterior: 21.3972). A Beta q can equal the posterior Beta(22, 16), so the gap can reach 0.

3 Reparameterise so the gradient passes through the draw



blue: parameters · red: random nodes · dashed: gradient path

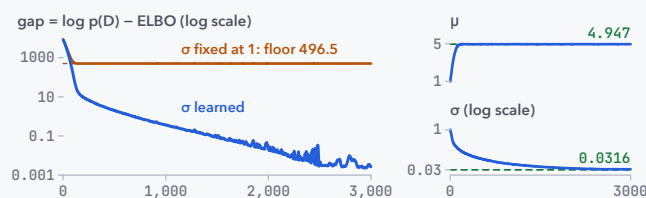
The ELBO is an expectation under q_ψ , and ψ is what we tune. Write each draw as a function of ψ and noise that has no parameters:

$$\theta = \mu + \sigma \epsilon, \epsilon \sim \mathcal{N}(0, 1) \Rightarrow \nabla_\psi \mathbb{E}_q[f(\theta)] = \mathbb{E}_\epsilon [\nabla_\psi f(\mu + \sigma \epsilon)]$$

and average over S draws. With $g = \partial \log p(D, \theta) / \partial \theta$: $\nabla_\mu \text{ELBO} = \mathbb{E}[g]$, $\nabla_\sigma \text{ELBO} = \mathbb{E}[g\epsilon] + 1/\sigma$ (the entropy term). In torch `q.sample()` returns a draw with no gradient (the notebook's error); `q.rsample()` applies the trick. The score-function estimator $\mathbb{E}_q[f \nabla_\psi \log q]$ is unbiased too, but much noisier:



4 Stochastic gradient ascent closes the gap



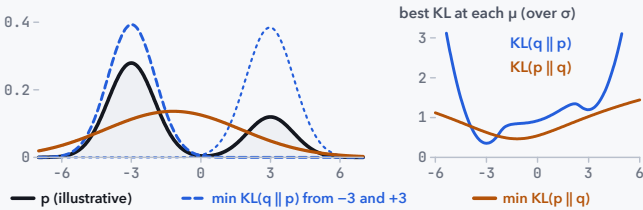
prior $\mathcal{N}(0, 1)$, 1000 draws from $\mathcal{N}(5, 1)$, seed 0 · Adam, 10 draws a step · green: exact posterior

The notebook's Gaussian mean model: prior $\mathcal{N}(0, 1)$, 1000 draws $x_i \sim \mathcal{N}(\theta, 1)$, $q = \mathcal{N}(\mu, \sigma^2)$ started at $\mathcal{N}(1, 1)$. The exact posterior $\mathcal{N}(\sum x_i / 1001, 1/1001) = \mathcal{N}(4.947, 0.0316^2)$ is the answer to hit. Each step uses 10 draws of ϵ and the gradients of panel 3. The gap falls from 8294 nats to 0.003 by step 3000, where $q = \mathcal{N}(4.948, 0.0331^2)$. The notebook keeps $\sigma = 1$ fixed: then the gap can never fall below 496.5, and its printed loss keeps jumping. Laplace would be exact here too, since this posterior is Gaussian.

Which KL, which family, and amortisation

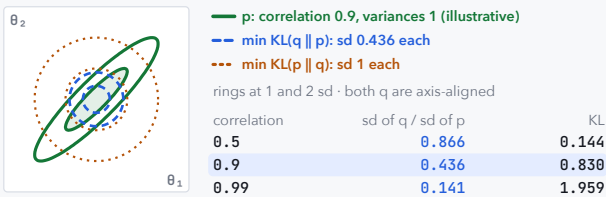
Reverse KL locks onto a mode, a factorised q misses correlations, and an encoder can output q for every data point.

5 Reverse KL seeks a mode, forward KL covers the mass



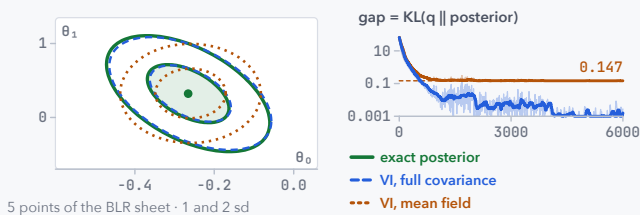
For $p = 0.7\mathcal{N}(-3, 1) + 0.3\mathcal{N}(3, 1)$, $\text{KL}(q \parallel p)$ is large wherever q has mass and p has almost none, so its best q hides in one peak: $\mathcal{N}(-2.99, 1.01^2)$ (KL 0.354), or from +3 the smaller one (1.197). $\text{KL}(p \parallel q)$ punishes $q \approx 0$ where p has mass, so it matches the mean and variance of p : $\mathcal{N}(-1.2, 2.93^2)$. VI uses the reverse KL, which needs expectations under q only. For the notebook's overlapping mixture the two nearly agree: (0.43, 0.72), (0.40, 0.75).

6 Mean field underestimates marginal variances



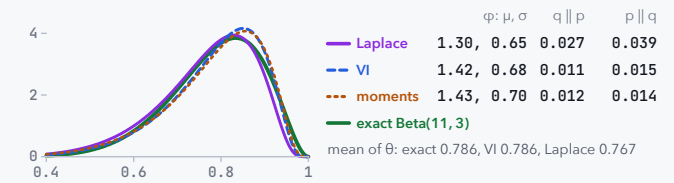
Mean field sets $q(\theta) = \prod_i q_i(\theta_i)$. For $p = \mathcal{N}(m, \Sigma)$, $\Lambda = \Sigma^{-1}$, the best factors under $\text{KL}(q \parallel p)$ are $q_i = \mathcal{N}(m_i, 1/\Lambda_{ii})$: the variance given all the other θ_j , at most the marginal Σ_{ii} . The sd shrinks by $\sqrt{1 - \rho^2}$; the means are right, the tilt is lost.

7 VI for Bayesian linear regression



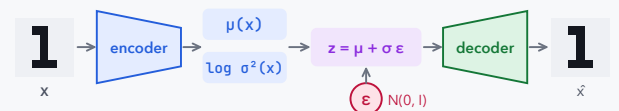
Prior $\mathcal{N}(0, 0.5I)$, noise sd 0.2: exact sds (0.103, 0.392), correlation -0.50 . Full covariance $q = \mathcal{N}(m, LL^T)$ reaches (0.102, 0.382), gap 0.003; mean field (0.088, 0.335), against $1/\sqrt{\Lambda_{ii}} = (0.089, 0.339)$ and gap $-\frac{1}{2} \log(1 - \rho^2) = 0.147$. The notebook's x are symmetric about 0, so there mean field is exact: sds 0.0705, 0.0244 (notebook fit 0.070, 0.024).

8 VI and Laplace on the same skewed coin



Both are Gaussians on the log-odds φ ; on θ itself a Gaussian puts mass where $p = 0$ and $\text{KL}(q \parallel p) = \infty$. Laplace copies the curvature at the peak. VI weighs the whole curve: the mean moves from 1.30 to 1.42 and the KL falls from 0.027 to 0.011. Its optimum has $\mathbb{E}_q[\partial_\varphi \log p] = 0$, here $\mathbb{E}_q[\theta] = 11/14$, the exact posterior mean.

9 A VAE is amortised VI



loss = reconstruction($x, \hat{x}$) + $\beta \cdot \text{KL}(\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, I))$
one network gives (μ, σ) for any x , instead of a separate (μ, σ) per data point

Each image x has a latent $z \sim \mathcal{N}(0, I)$ and a decoder $p(x \mid z)$. One encoder outputs $\mu(x), \log \sigma^2(x)$ for every image; $z = \mu + \sigma \epsilon$ carries the gradient back:

$$\text{ELBO}(x) = \mathbb{E}_q[\log p(x \mid z)] - \frac{1}{2} \sum_j (\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2)$$

The notebook (quoted): 1000 MNIST images, latent size 2 to 256, KL weight β from 0 to 1. In 2D two zeros sit 0.838 apart at $\beta = 0$, 0.0022 at $\beta = 1$: the KL pulls codes onto the prior. It averages over pixels and dimensions, so its $\beta = 1$ is not the ELBO.

10 Pause and think

Why not minimise $\text{KL}(p \parallel q)$? It needs draws from the posterior.

Higher ELBO, better q ? Within one model, yes; across models the gaps differ.

VI or MCMC? VI is fast but limited by the family; MCMC is exact in the limit, and slow.

Why do printed losses jump? Each is a Monte Carlo estimate from a few draws. Watch the parameters, or the exact ELBO when you have it.

Is the ELBO a bound on the likelihood of θ ? No: on $\log p(D)$, and it is tight only when q is the posterior.

log p(D) = ELBO + KL(q || posterior) · raise the ELBO · sample $\theta = \mu + \sigma \epsilon$ · reverse KL: one mode, too narrow

Source: slides "Variational inference" and the variational inference and VAE notebooks, nipunbatra.github.io/pml-teaching. Numbers recomputed in NumPy (closed forms, grids, Gauss-Hermite quadrature, seeded Adam runs); torch and MNIST outputs are quoted. Fixes: the notebook labels its Adam prints one block late ("iteration 200" is after 300 steps); the manual reparameterisation cell takes $\log q$ from a stale q ; the full-covariance regression sums $\log q$ over its 100 draws, so its covariance comes out about 100 times too large, and it prints LL^T without the triangular mask; its data have noise sd 5 while the model assumes 1; the loss 218.9 for $q = \text{Beta}(0.1, 2)$ is a 100-draw estimate of 232.9. The slides' first KL exercise frame answers in μ_1, σ_1 notation.